

Introduction of Foundations and Basements

Introduction to Foundations:

- A foundation is the lowest part of a building structure that transfers the load of the building safely to the soil below.
- It provides stability, prevents settlement, and ensures the safety and durability of the structure.

Introduction to Basements:

- A basement is a floor or part of a building constructed below ground level.
- Basements are commonly used for parking, storage, mechanical rooms, commercial spaces, or additional living areas.

Functions of Basements:

- Provide additional usable space
- Improve structural stability
- Protect utilities and mechanical equipment
- Offer parking or storage facilities
- Help in temperature insulation

Soil (site) Exploration:

Introduction:

- Soil exploration is the process of investigating the soil and subsoil conditions at a site to determine their engineering properties for safe and economical design of foundations.
- The main objective of soil exploration is to know the general picture of the geography of that area.
- To get adequate information of type and nature of soil available at different depths for designing safe, sound and economical foundation for a structure.

Purposes of Soil Exploration:

- To determine the value of soil bearing capacity.
- To select economical type of foundations.
- To fix the depth of proposed foundation.
- To predict the probable settlement of foundation and to make the necessary allowance in the design.
- To know the underground water level.
- To know suitability of soil and degree of compaction of soil.

Factors to be Considered Before Planning Site Investigation:

- Size of structure.
- Depth of basement.
- Height of structure.
- Loads to be transmitted by the foundation.
- Spacing of columns and load bearing walls.

Methods of Soil (Site) Exploration:

Soil exploration is carried out by:

1. Inspection of Site

Following information are gathered during inspection of site:

- Hard, soft, water logged, etc.
- Classification of soil by visual examination such as gravel, sand, clay.
- Behavior of ground during changes in ground water level, drainage conditions.
- Unstable sites, slopes, etc.

2. Probing

A steel bar of 25 to 40 mm in diameter hammered into the soil until the hard sub-structure is met and nature of the soil is obtained by examining the soil sticking to the sides of the bar.

3. Means of test pit

- Holes dig for the entry of person for inspection of soil at random locations are called test pits.
- Test pits are square in plan and depths depend on the foundation type.
- This method is suitable for depth up to 3m.
- Suitable for cohesive soil.

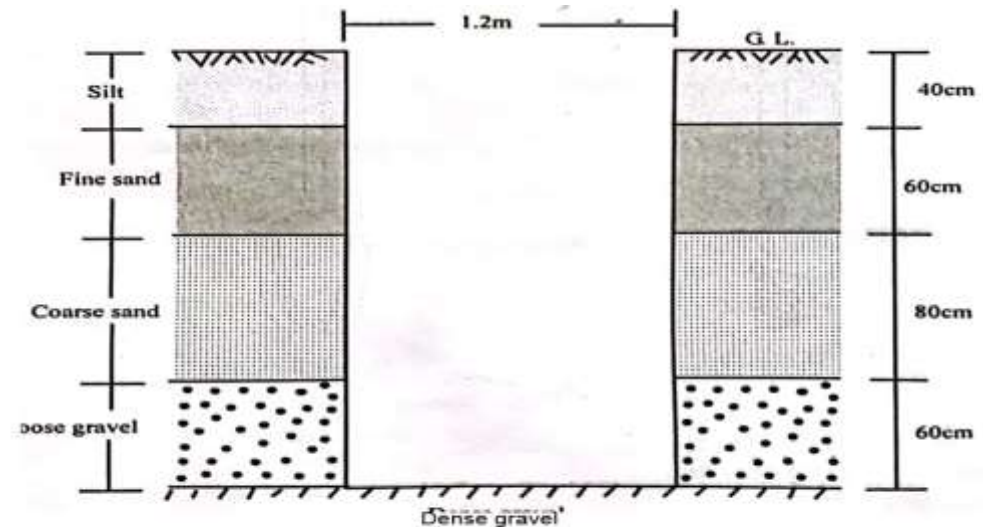


Fig.2.1: Test Pit.

4. Boring

In this method boring into soil extracting soil samples at various depths and conducting laboratory test is carried out to determine soil characteristics especially the strength.

Different method of boring are:

- (a) Auger boring
- (b) Wash boring
- (c) Rotary drilling
- (d) Shell and auger boring

(a) Auger Borings

- Simplest method of exploration and sampling.
- Power driven or hand operated.
- Maximum depth 10 m.
- Suitable in all soils above GWT but only in cohesive soil below GWT.
- Hollow stem augers used for sampling or conducting standard penetration tests.



Helical auger

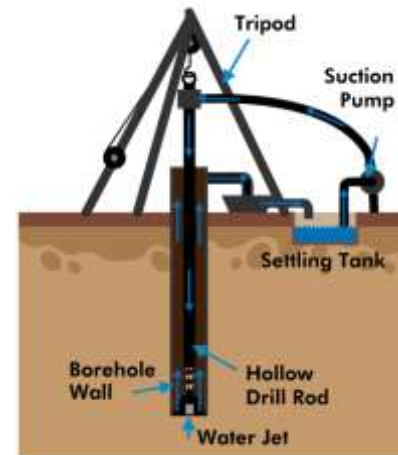


Post-hole auger

Fig. 2.2: Auger Boring.

(b) Wash Boring

- Wash boring is a method of advancing boreholes for site exploration and geotechnical investigation, using a jet of water to flush soil cuttings out of the borehole, particularly effective in waterlogged or sandy soils
- A hollow drill rod with a sharp chisel or chopping bit at the lower end is inserted into the casing. Water is forced through the drill rod, and the resulting chopping and jetting action disintegrates the soil.



Foundations and Their Types

- A foundation is that part of structure which is in direct contact with the ground to which the loads are transmitted.

□ Every building consists of two basic components

- The super structure.
- The sub-structure(Foundation).

❖ Superstructure

- Superstructure is usually that part of the building which is above ground.

❖ Sub-structure

- Substructure or foundation is the lower part of the building usually located below ground level, which transmits the loads of the superstructure to the soil.

Functions of Foundation:

Transmits the superimposed loads (wind, vibration, dead load and live loads).

Prevent against settlement.

Helps to distribute loads of superstructure evenly.

Protect against soil movement.

Provides lateral stability.

Characteristics of Foundation:

- Foundation should be deep enough to increase stability and prevent overturning.
- Foundation base should be rigid to minimize settlements especially for the case when super-imposed loads are not evenly distributed.
- Foundation should be wide enough so as to distribute the weight of the super-structure over large area.

Selection of Foundation:

The selection of the foundation type for a particular site depends on the following considerations:

- Nature of subsoil.
- Nature and extent of difficulties
- Availability of expertise and equipment.

Types of Foundation

1. Shallow Foundation
2. Deep Foundation

1. Shallow Foundation

- This is the most common type of foundation and can be laid using open excavation by allowing natural slopes on both sides.
- Depth of foundation is less than or equal to width.
- It transmits superimposed load to the soil at a level close to the ground floor of the building.

Types of Shallow Foundation:

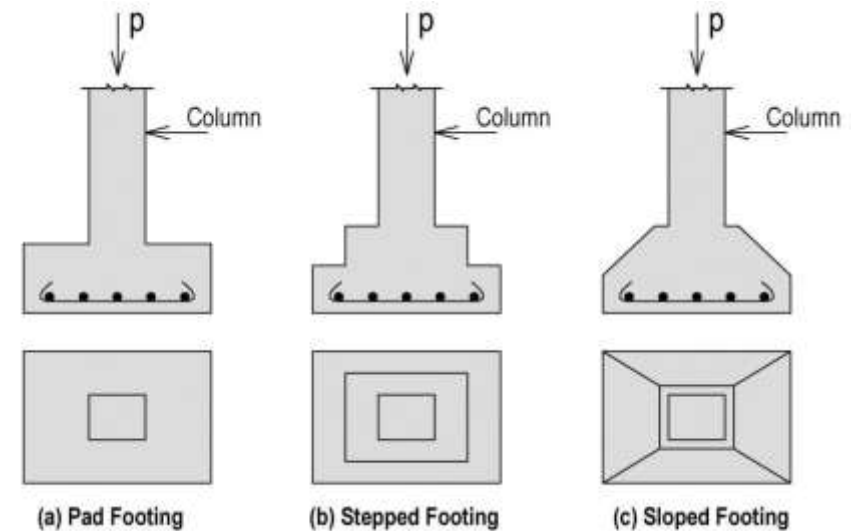
(a) Spread Footing or Pad Foundation

Spread footing is the one which supports either one column or one wall. The various types of spread footings are:

- (i) Isolated or column footings.
- (ii) Combined footings.
- (iii) Strip footing.
- (iv) Mat (Raft) Foundation

(i) Isolated or Column Footings:

- They are used to support individual columns.
- In case of heavy loaded columns, steel reinforcements are provided.
- Generally, 15cm offset is provided on all sides of concrete bed.
- The footing of concrete columns may be slab, stepped or sloped type.

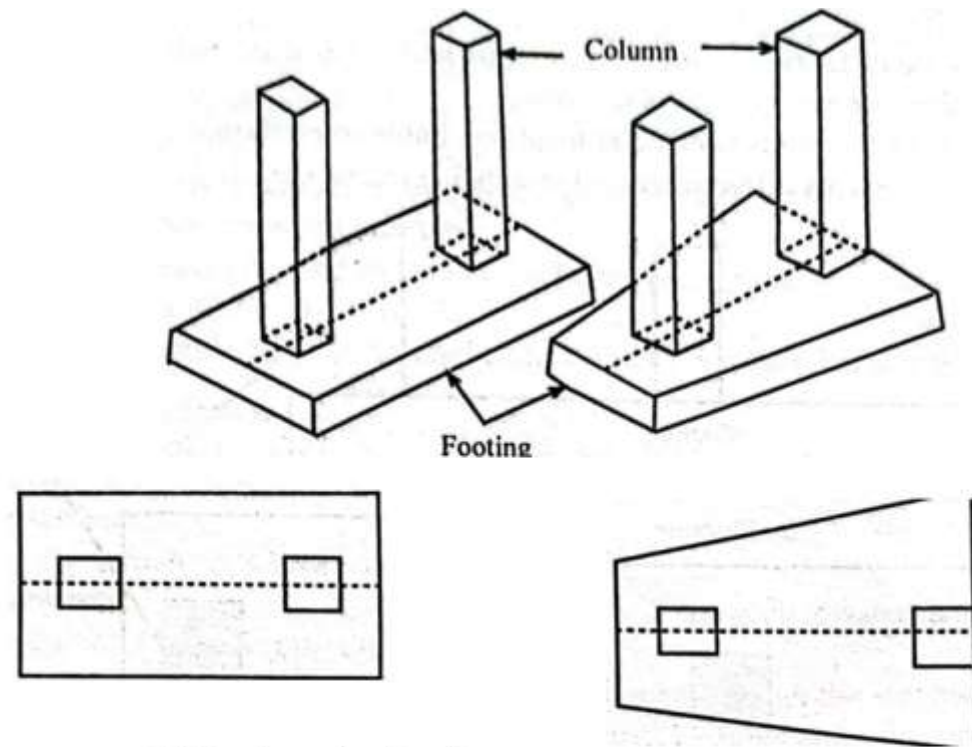


(ii) Combined Footings

- A combined footing supports two or more columns in a row.
- The combined footing can be rectangular in shape if both the columns carry equal loads or can be trapezoidal if both the loads are unequal.
- Generally, they are constructed of reinforced concrete.

This type of footing is provided under the following circumstances:

- When columns are very near to each other and their individual footings overlap.
- When bearing capacity of the soil is less, requiring more area under individual footing.



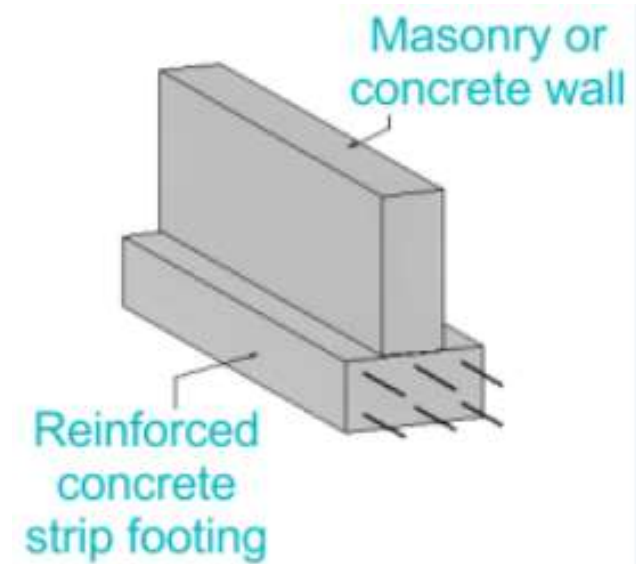
(b) Rectangular Footing.

(c) Trapezoidal Footing.

Fig.2.4: Combined Footing.

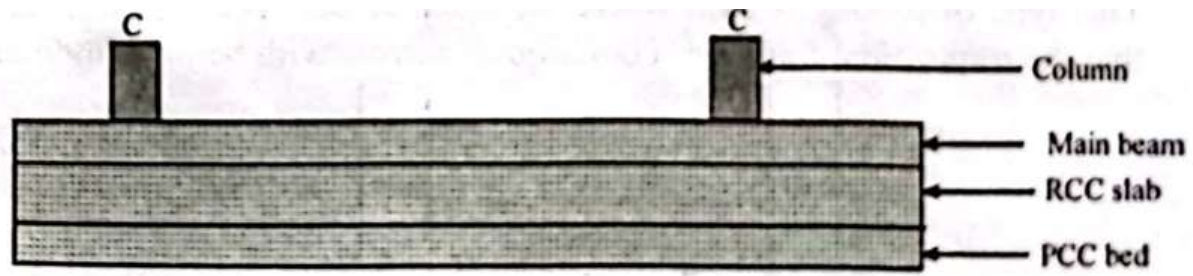
(iii) Strip Foundation

- A strip foundation is a type of shallow foundation used to support load-bearing walls.
- It consists of a continuous strip of concrete laid under walls to distribute the load evenly to the soil.
- Suitable for (Residential buildings, Boundary walls)

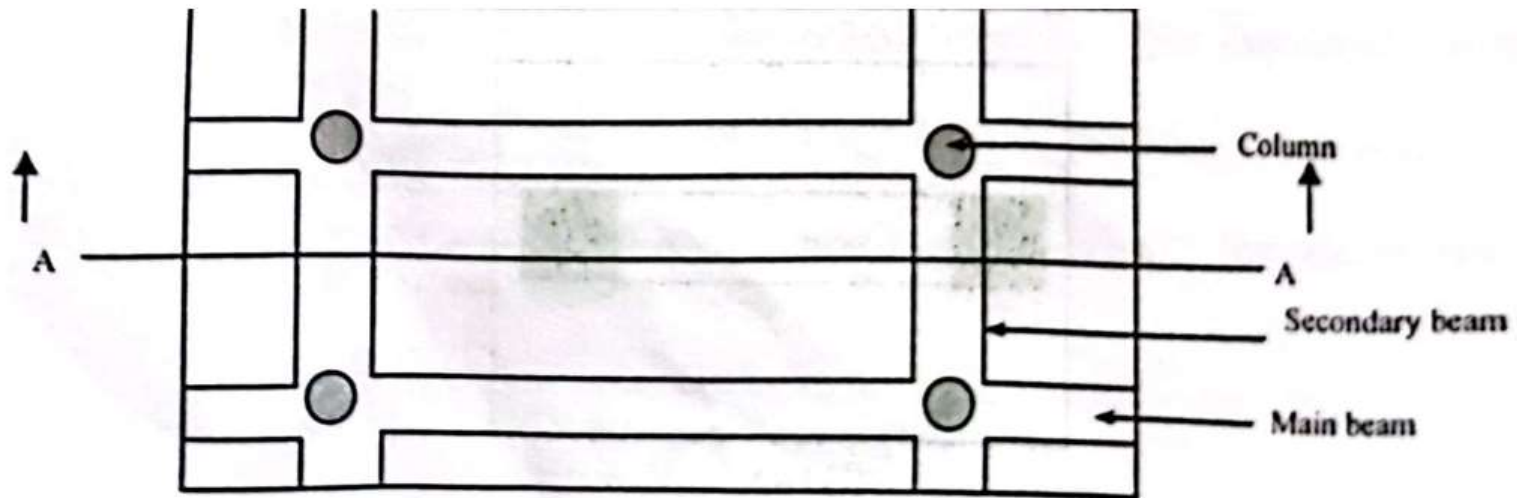


(iv) Mat (Raft) Foundation

- A raft foundation is a combined footing that covers the entire area beneath a structure and support all the walls and columns.
- It is a flat concrete slab, heavily reinforced with steel, which carries the downward loads of the individual columns or walls.
- Raft foundations are used to spread the load from a structure over a large area, normally the entire area of the structure.



(a) Section along A-A



(b) Plan

Fig.2.7:Mat Foundation.

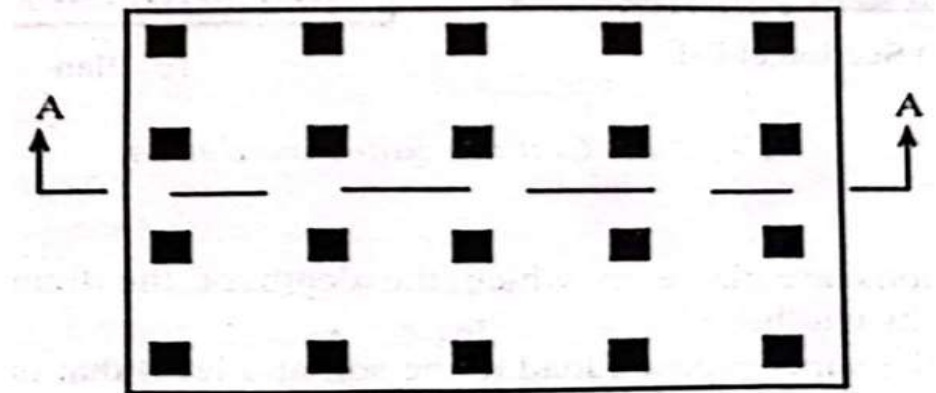
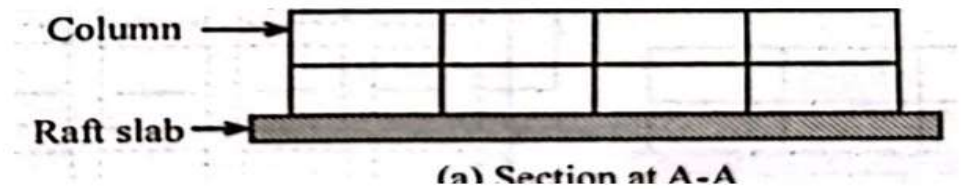
Use of Mat Foundation

- When an allowable soil pressure is low the building loads are heavy mat foundation is economical to use.
- Use to reduce settlement.
- The bottom of the structure is located below the groundwater table, so waterproofing is an important concern.
- The lateral loads are not uniformly distributed through the structure and thus may cause differential horizontal movements in spread footings and pile caps. The continuity of a mat will resist such movement.

Types of Raft (Mat) Foundation:

(i) Solid Raft

- Is a solid slab of concrete reinforced in both directions and used for light framed structures or small buildings.
- Used for slab thickness up to 300mm.



(b) Plan (Solid Raft Foundation.)

(ii) Slab and Beam Foundation

- Used for slab thickness greater than 300mm in this case it is cheaper to use slab and beam construction.

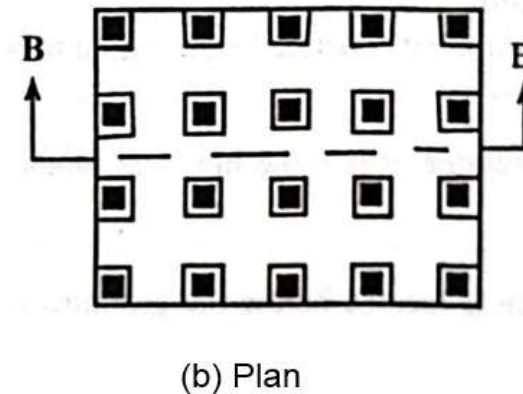
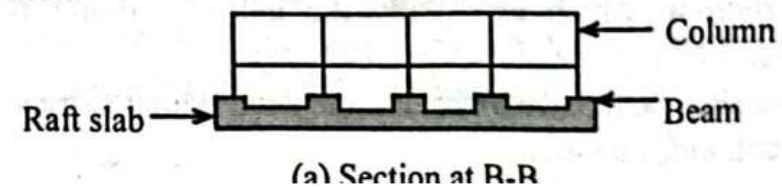
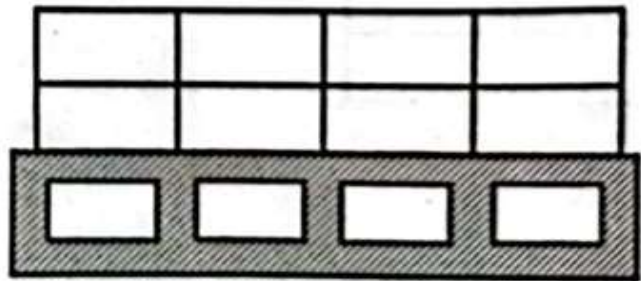


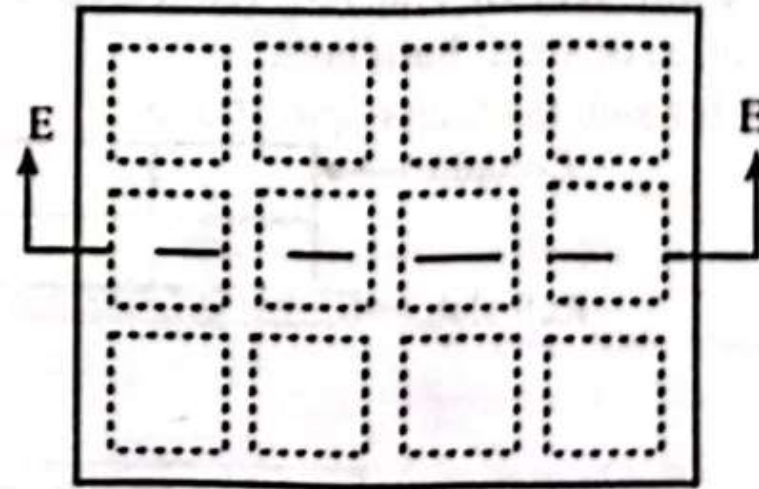
Fig.2.9:Slab and Beam Foundation.

(iii) Cellular Raft

- Used where stresses in the raft are high and great rigidity is required.
- Beam is deep and overall raft is likely to exceed 900mm.



(a) Section at E-E



(b) Plan

Fig.2.10: Cellular Raft Foundation.

2. Deep Foundation

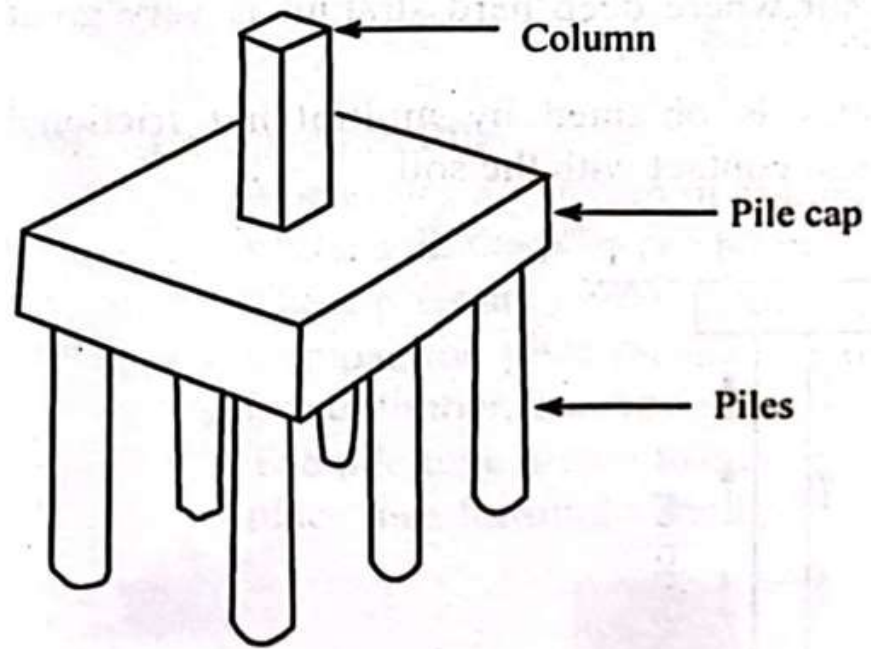
- The depth of the foundation is very large in comparison to its width.
- They transfer the superimposed load to the soil at a level that is at a great distance from the lowest floor.
- These foundations carry loads from a structure through weak incompressible soils to the stronger and less compressible soils at depth.

Types of Deep Foundations

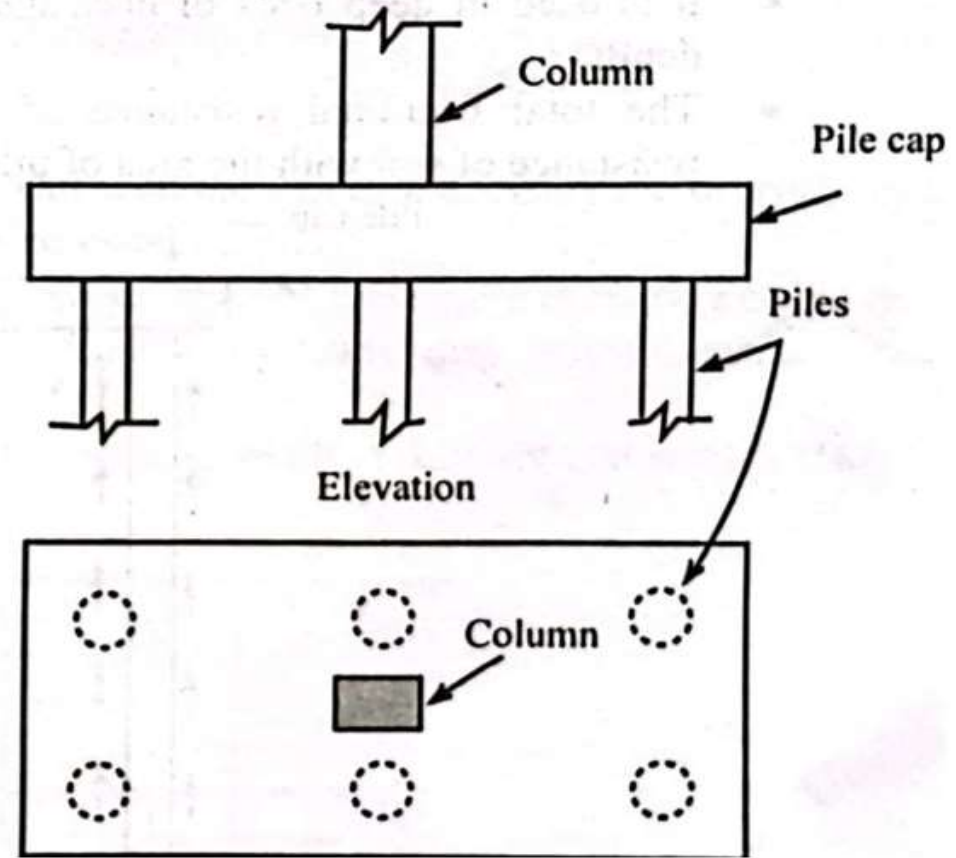
- (a) Pile foundation
- (b) Pier foundation
- (c) Well or caisson foundation

(a)Pile Foundation

- It is that type of deep foundation in which loads are taken to low level by means of vertical members which may be timber, concrete or steel.
- It may be used instead of raft foundation where no firm bearing strata exists at a reasonable depth and the loading is uneven.
- The piles may be placed separately or they may be placed in form of a cluster throughout the structure.



(a) Isometric View of Pile Foundation



(b) Plan

Uses of Pile Foundation

- When load of super structure is heavy and its distribution is uneven.
- The top soil has poor bearing capacity.
- When structure is situated on the sea shore or river bed.
- Water table is high.
- Canal or deep drainage lies near the foundation.

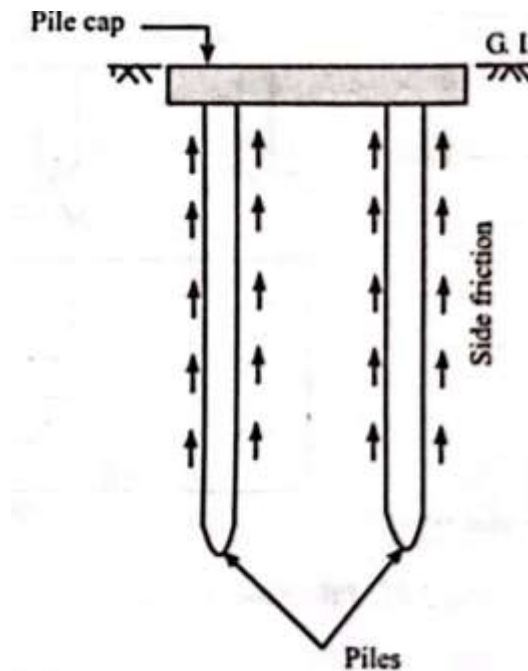
Classification of Pile Foundation

A. Types of Piles According to Uses

- (i) Friction piles
- (ii) End bearing piles
- (iii) Compaction piles
- (iv) Tension pile or uplift pile
- (v) Batter pile
- (vi) Sheet pile

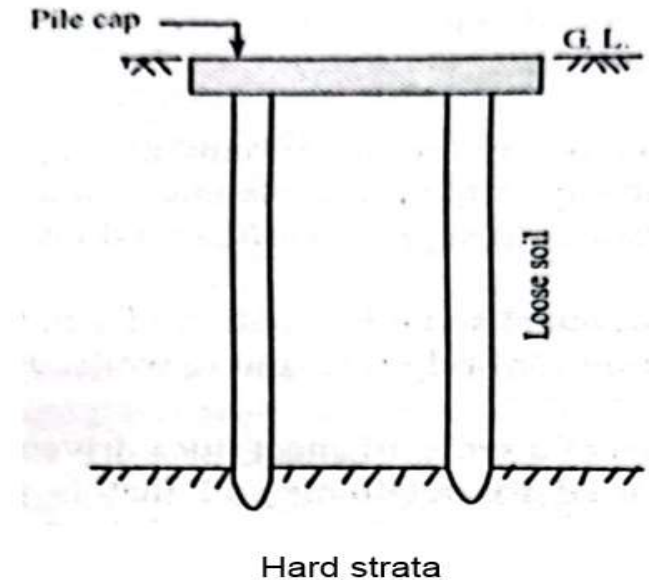
(i) Friction Piles

- They are used to transfer load to the surrounding soil by means of the friction between their surfaces and the soil.
- It is used in deep beds of clay and silt where deep hard stratum is very greater depth.



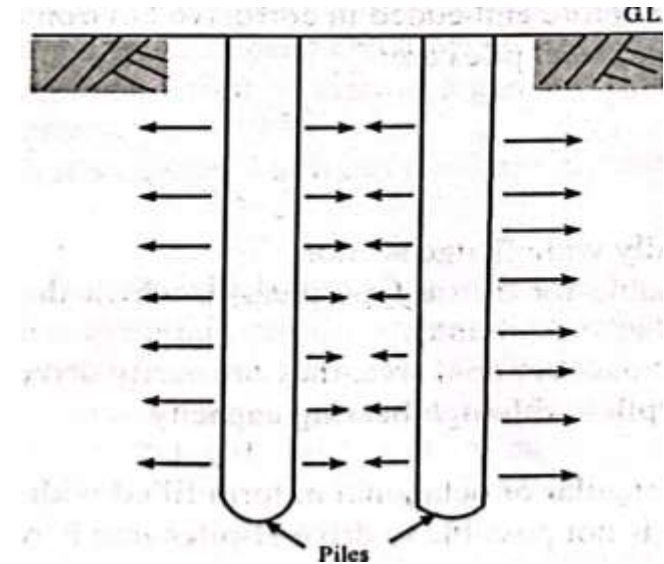
(ii) End Bearing Piles

- These are ordinary piles, driven vertically to take up vertical loads.
- They are driven up to the hard stratum and transmit the load to the hard stratum.
- Helps to transmit the loads to a greater depth.



(iii) Compaction Pile

- When piles are driven in granular soil with the aim of increasing the bearing capacity of the soil, the piles are termed as compaction piles.
- These piles are used to compact loose soil thus increasing their bearing capacity.
- Compaction piles themselves do not carry any loads, they may be weaker material (timber, bamboo, etc.)



(iv) Tension Pile or Uplift Piles

- Piles supporting high structures such as transmission towers, water towers are subjected to uplift forces due to wind.
- Overturning caused by wind, ice loads and broken wires may produce large tension forces for power transmission towers.
- These piles anchor down the structure subjected to uplift pressure due to overturning movement.

(v) Batter Piles

- Piles are driven vertically into the ground are capable of supporting vertical loads but are not able to support appreciable lateral loads.
- In such circumstances to support vertical and inclined loads batter piles are used.
- These piles are used for bridges retaining walls, abutments etc.

(vi) Sheet Pile

- Sheet pile consists of a series of sheet piles driven side by side into the ground.
- These piles are generally used for temporary construction, light weigh construction, coffer dams, river bank protection and retaining the sides of excavation.

B. According to Material Use:

(i) Steel Pile

- They are usually driven with open ends into the soil.
- A conical tip is used where the piles have to penetrate boulders & rocks.
- It comes in various shapes and sizes.
- It needs to be treated before embedded in corrosive environment.

(ii) Cement Concrete Piles

- Usually have square/circular/octagonal cross sections.
- It has higher capacity than timber piles.

Cement concrete piles may be classified as:

a) Pre-cast piles

- They are reinforced concrete piles and are moulded in circular, square, rectangular or octagonal cross sections.
- They are cast and cured in casting yard and transported to the construction site.

b) Cast-in-situ Piles

- They are RCC piles which are constructed or casted in the place where they are to be used finally.

(iii) Timber Piles

- Have been used since ancient times.
- The timber piles may be circular or square.
- Even though it's cheaper but it has low capacity and can't take hard driving.

(iv) Composite Piles

- This is a type of pile which is made of two or more than two materials.
- Two different materials are driven one over other so as to enable them to act together to perform the function of a single pile.

(c) Caisson or Well Foundation

- These are the box like structures circular or rectangle in plans which are sunk from the surface of either land or water to the desired depth.
- The term caisson has been derived from the French word 'CAISSEE', meaning BOX.
- It is commonly used where foundation under water is done.

Element of Caisson Foundation

1.Bottom Plug

- Transfer the load of superstructure from well to the soil below.
- It casts under water after the final depth of well is reached.
- No reinforcement is provided.

2. Sand Filing

- Sand filling will commence after a period of 3 days laying of bottom plug.
- It helps in increasing the weight of foundation by providing more stability to the structure.

3. Intermediate Plug

- The function of the plug is to keep the sand filling sandwiched.

4. Top Plug

- After filling up to the required level a plug of concrete is provided over it.
- It also serves as shuttering for laying of well cap.

5. Well Cap

- It is RCC slab.
- It is generally kept at low water level.

6. Steinning

- It is designed such that it can be sunk under its own weight.

7. Well Curb

- It is made of RCC with steel cutting edge.

8. Cutting Edge

- It is either projected below the curb as sharp edge or can also have flat bottom.
- It is a steel member.

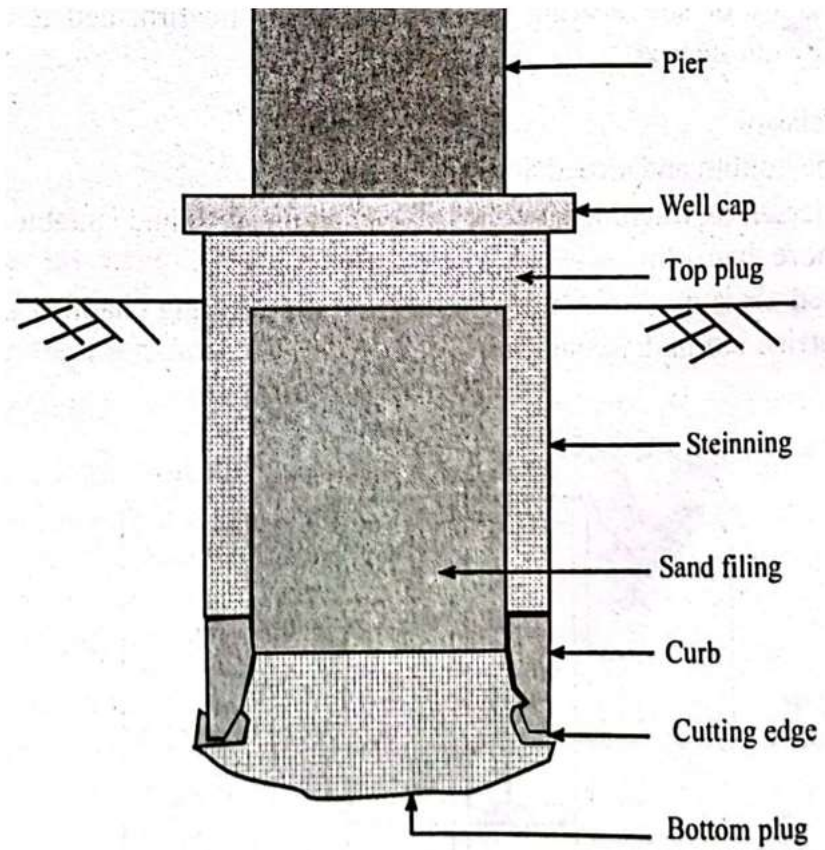


Fig: Caisson Foundation.

Types of Caisson

(i) Box Caisson

- Also called as well caisson.
- It is a strong water tight vessel built of timber, reinforced concrete or steel.
- This type of caisson is suitable where loads are not very heavy.

(ii) Open Caisson

- Open caisson is open both at top and bottom.
- Built of timber, steel or reinforced concrete, etc.
- Used on sandy or soft bearing stratum and where no firm bed is available for large depth below the surface.

(iii) Pneumatic Caisson

- Open at the bottom and closed at top.
- Useful at locations where it is not possible to adopt wells and suitable when the depth of water is more than 12m.

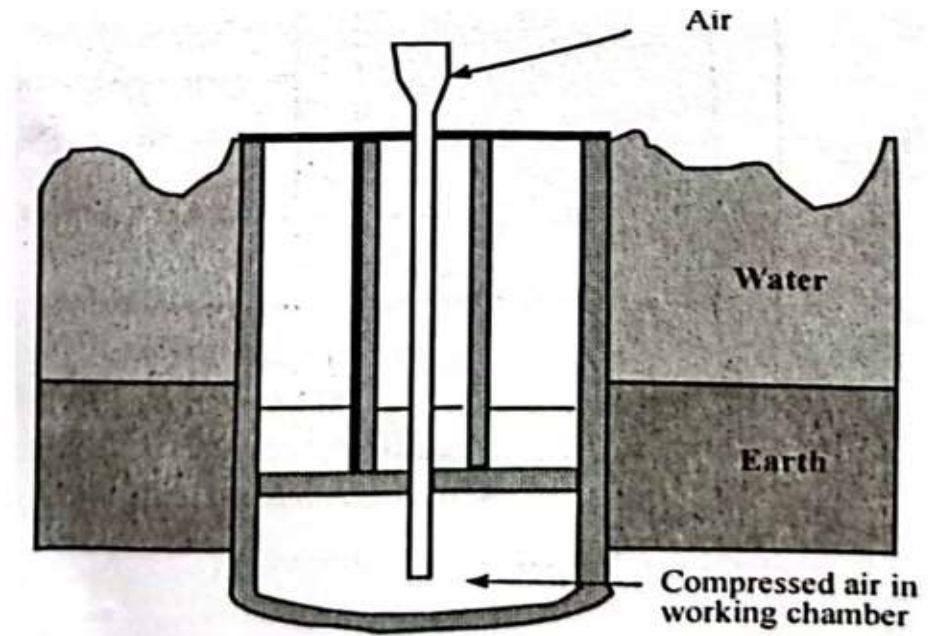


Fig.2.16: Pneumatic Caisson.

Earthwork in Excavation of Foundations:

Site Clearance:

- Before foundation is to be laid it is necessary to excavate the trench of required depth and width.
- It is the process of clearing the site (where construction work is to be carried out) of any existing structure, truss and any kind of disturbing elements which may obstruct the construction work.

Site Clearance Work Includes:

- Removal of top soil.
- Removal of trees.
- Demolition of existing structure.

Earthwork in Excavation:

- Excavation in foundation is done either manually with the help of conventional implements or mechanically or both together depending on factors such as soil conditions, shape of cut section, ground water and time required to complete the job.
- Excavation works include foundations, basements, trenches, etc.

Conventional implement are:

- Spade
- Pick axe
- Rammer etc

Excavating equipment are:

- Drag shovel (excavator, backhoe loader)
- Multi bucket trencher, etc.

Excavation in Soft Soil

- In the case of excavation in soft soil the poling boards are placed closer say 90 cm apart and are held by wailings.
- In loose, wet or sandy soils, the sides of trenches sometimes may be supported by open timbering, close-boarded timbering or sheeting.

Excavation in Hard Soil

- In the case of excavation in hard soils where depth of excavation is moderate the sides of trench required no support.
- If the depth of hard soil is more than 1.5 m deep should be properly shored and timbered.

Wet Excavation

- Excavation in foundation trenches in ground having high water table or in water logged area possess great problems.
- There are various methods to deal with this situation depending upon the depth of excavation, depth of water table and other factors.

Following methods are adopted while excavating foundation in water logged area:

- By constructing drain.
- By constructing deep well.
- Well point system.
- Vacuum method.
- By chemical consolidation.
- Freezing process.

Waterproofing Techniques in Basements:

- Basement waterproofing is the process of preventing water seepage and moisture penetration into the basement of a building.
- Proper waterproofing increases the durability, safety, and service life of the structure.

Objectives of Basement Waterproofing

- Prevent leakage and dampness
- Protect structural members from corrosion
- Avoid fungal growth and unhealthy conditions
- Increase durability of the building
- Protect stored materials and equipment

Common Waterproofing Techniques in Basements:

1. Cementitious Waterproofing

- Uses cement-based waterproof coatings
- Applied on walls and floors using brush or trowel
- Simple and economical method
- Suitable for internal waterproofing

Advantages

- Easy to apply
- Low cost

Disadvantages

- Less flexible
- May crack due to movement

2. Bituminous Coating

- Also called asphalt coating
- Applied as a protective layer on basement walls

Types

- Hot applied bitumen
- Cold applied bitumen

Limitation

- Can get damaged by sunlight and high temperature

3. Bituminous Membrane Waterproofing

- Uses rolls of waterproof membrane sheets
- Membrane is laid on basement walls and slabs

Suitable For:

- Foundations
- Retaining walls
- Basements with high water pressure

4. Liquid Waterproofing Membrane

- Liquid coating forms a rubber-like waterproof layer after drying

Materials Used

- Polyurethane
- Polymer-based compounds

Advantages

- Flexible and crack resistant

5. Crystalline Waterproofing

- Special chemicals react with concrete and form crystals
- Crystals block pores and capillaries in concrete

Advantages

- Self-sealing property
- Long-lasting protection
- Suitable for underground structures

6. Polyurethane Waterproofing

- Highly flexible waterproof coating
- Used in areas exposed to heavy water seepage

Advantages

- High durability
- Strong water resistance

Disadvantages

- Expensive
- Surface preparation is important

7. Drainage System Waterproofing

- A drainage system is installed around the basement to remove water.

Purpose

- Reduces hydrostatic pressure around foundation walls

External Waterproofing Techniques:

- These methods are applied outside the basement wall.

Common External Methods

- Membrane waterproofing
- Protective drainage boards
- Exterior drainage pipes

Benefits

- Stops water before entering the structure
- Provides long-term protection

Internal Waterproofing Techniques

- Applied inside the basement to control moisture.

Methods

- Interior sealants
- Waterproof paints

Suitable For

- Minor leakage problems
- Existing buildings

Basement Waterproofing Materials

- Common materials used:
- Bitumen
- Cement slurry
- PVC membranes
- Polyurethane compounds
- Epoxy resins
- Bentonite sheets

Factors Affecting Selection of Waterproofing Method

- Groundwater level
- Soil condition
- Type of structure
- Water pressure
- Cost and maintenance
- Climate condition

Thank You